

Quantum Phase Estimation for a Controlled-Phase Gate

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Abstract

This project implements Quantum Phase Estimation (QPE) in Qiskit for the unitary $U = CP(\phi)$ and the eigenvector $|11\rangle$. The implementation estimates $\theta = \phi/(2\pi)$, validates exact and non-exact binary phases, studies the effect of counting-register precision, and evaluates robustness under a simple noise model. The work is deliberately positioned between a minimal demonstration and a hardware-scale study: it contains a reusable circuit implementation, quantitative experiments, complexity discussion, and reproducible Colab workflow.

1 Problem Statement

The selected exam project requires implementation of QPE for a controlled-phase gate CR_ϕ with eigenvector $|11\rangle$. Since

$$CP(\phi)|11\rangle = e^{i\phi}|11\rangle = e^{2\pi i\theta}|11\rangle, \quad \theta = \frac{\phi}{2\pi}, \quad (1)$$

the task is to recover an m -bit approximation to θ . This directly follows the project description supplied for the course.

2 Background

QPE is a central primitive in quantum algorithms. Given an eigenstate $|u\rangle$ of a unitary U , it transfers phase information from repeated controlled powers U^{2^j} to a counting register. An inverse Quantum Fourier Transform (QFT) converts the relative phases into a binary estimate [1, 2].

3 Circuit Design

The circuit uses m counting qubits and two target qubits. The target register is initialized to $|11\rangle$. Hadamard gates create a uniform superposition in the counting register. For counting qubit j , the controlled operation is $U^{2^j} = CP(2^j\phi)$. Because U already has one control, adding the QPE control produces a phase on $|11\rangle$, implemented as a two-control phase gate. Finally, an inverse QFT and measurement produce the estimate.

Listing 1: Core controlled-power construction

```
for j in range(precision):
    gate = MCPhaseGate(phi * (2 ** j), num_ctrl_qubits=2)
    qc.append(gate, [count[j], target[0], target[1]])
```

4 Experimental Method

All experiments use Qiskit Aer with fixed seeds and 4096–8192 shots. Four phases are tested: $1/8$, $3/8$, $5/16$, and 0.30 . The first three are exactly representable with a sufficiently large binary register; the final value is not. The modal measurement string is decoded as

$$\tilde{\theta} = \frac{k}{2^m}, \quad (2)$$

where k is the measured integer. Circular error is used because phase is periodic.

5 Results and Interpretation

For exactly representable phases, the ideal simulator produces a dominant bit string corresponding to the correct binary fraction. For $\theta = 0.30$, probability is distributed around nearby grid points $k/2^m$. Increasing m reduces quantization error, although the probability of a single modal string need not increase monotonically because the exact phase remains between grid points.

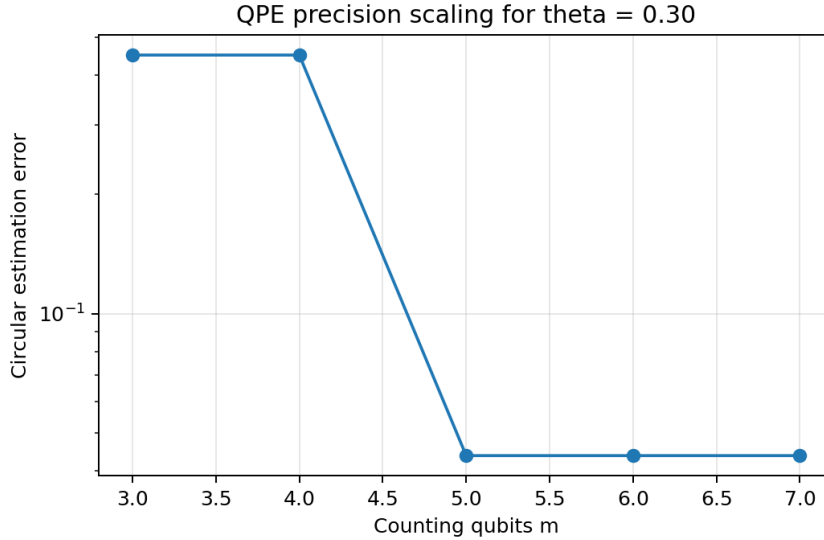


Figure 1: Estimation error versus number of counting qubits.

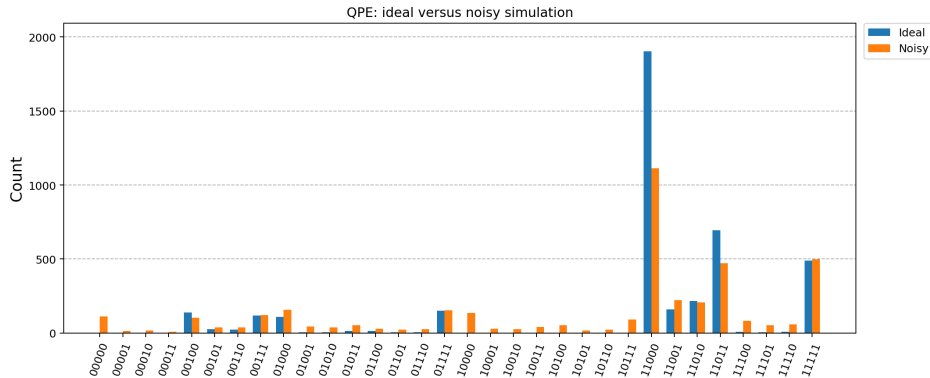


Figure 2: Ideal and noisy output distributions for an exactly representable phase.

6 Noise Study

A pedagogical noise model combines depolarizing gate errors with symmetric readout error. The expected effect is a reduction of the correct peak and increased probability in neighboring or unrelated outcomes. This is not a calibrated model of a specific processor; it is a controlled experiment showing that QPE’s useful information is encoded in interference and is therefore sensitive to accumulated gate error.

7 Complexity

The direct inverse QFT uses $O(m^2)$ controlled rotations. At the algorithmic level there are m controlled powers of U . For this specific U , each controlled power is represented by a multi-controlled phase gate. On hardware, decomposition into the native basis increases two-qubit gate count and depth. Consequently, the high-level QPE circuit may look compact while its transpiled realization is substantially more expensive.

8 Limitations and Extensions

The study uses simulation and a simplified noise model. Extensions include approximate QFT, execution on an IBM backend, zero-noise extrapolation, and comparison of alternative decompositions of the multi-controlled phase operation.

9 Conclusion

The project demonstrates QPE for the required controlled-phase unitary and eigenstate, then extends the basic task through precision and noise experiments. Exact binary phases are recovered sharply in ideal simulation, non-exact phases show finite-grid spreading, and noise lowers peak confidence. The implementation is compact enough for a course project while retaining meaningful algorithmic and experimental analysis.

References

- [1] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 2010.
- [2] A. Yu. Kitaev. “Quantum measurements and the Abelian stabilizer problem.” arXiv:quant-ph/9511026, 1995.
- [3] IBM Quantum. *Qiskit Documentation*. Software documentation consulted for the implementation.